PHOTO ELECTRIC EFFECT

EXAMPLE 7.2

A radiation of wavelength 300 *nm* is incident on a silver surface. Will photoelectrons be observed?

Solution:

Energy of the incident photon is

E = hν = hc/λ (in joules)

E = hc/λe (in eV)

Substituting the known values, we get

$E = 6.626\times10^{-34} \times3\times10^{8} / 300\times10^{-9} \times1.6\times10^{-19}$

$E = 4.14\ eV$

From Table 7.1, the work function of silver = 4.7 *eV*. Since the energy of the incident photon is less than the work function of silver, photoelectrons are not observed in this case.

EXAMPLE 7.3

When light of wavelength 2200$\mathring{A}$ falls on Cu, photo electrons are emitted from it. Find (i) the threshold wavelength and (ii) the stopping potential. Given: the work function for Cu is $\phi_0 = 4.65\ eV$.

Solution

i) The threshold wavelength is given by

$$\lambda_0 = \frac{hc}{\phi_0} = \frac{6.626\times10^{-34}\times3\times10^{8}}{4.65\times1.6\times10^{-19}}$$

$$= 2672\mathring{A}$$

= 2672 $\mathring{A}$

ii) Energy of the photon of wavelength 2200 $\mathring{A}$ is

$$E = \frac{hc}{\lambda} = \frac{6.626\times10^{-34}\times3\times10^{8}}{2200\times10^{-10}}$$

$$= 9.035\times10^{-19} J = 5.65\, eV$$

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We know that kinetic energy of fastest photo electron is

$K_{max} = h\nu - \phi_0 = 5.65 - 4.65$

$= 1\ eV$

From equation (7.3), $K_{\max} = eV_0$

$V_0 = K_{\max}/e\ 1\times1.6\times10^{-19} / 1.6\times10^{-19}$

Therefore, stopping potential = $1\,V$

EXAMPLE 7.4

The work function of potassium is 2.30 eV. UV light of wavelength 3000 $\mathring{A}$ and intensity $2Wm^2$ is incident on the potassium surface. i) Determine the maximum kinetic energy of the photo electrons ii) If 40% of incident photons produce photo electrons, how many electrons are emitted per second if the area of the potassium surface is 2 cm^2 ?

Solution

i) The energy of the photon is

E = hc/λ

$= 6.626\times10^{-34} \times 3\times10^{8} / 3000\times10^{-10}$

$E = 6.626\times10^{-19}\, J = 4.14\ eV$

Maximum KE of the photoelectrons is

$K_{max} = h\nu - \phi_0 = 4.14 - 2.30 = 1.84\ eV$

ii) The number of photons reaching the surface per second is

$n_p = P/E\ \times A$

$= [\ 2 / 6.626\times10^{-19}\ ] \times [2\times10^{-4}]$

$= 6.04\times10^{14}$ photons / sec

The rate of emission of photoelectrons is

$= (0.40)n_p = 0.4\times6.04\times10^{14}$

$= 2.415´10^{14}$ photoelectrons / sec

EXAMPLE 7.5

Light of wavelength 390 *nm* is directed at a metal electrode. To find the energy of electrons ejected, an opposing potential difference is established between it and another electrode. The current of photoelectrons from one to the other is stopped completely when the potential difference is 1.10 *V*. Determine i) the work function of the metal and ii) the maximum wavelength of light that can eject electrons from this metal.

Solution

i) The work function is given by

$$\phi_0 = hv - K_{max} = \frac{hc}{\lambda} - eV_0$$

since $K_{max} = eV_0$

$$= \left[\frac{6.626\times10^{-34}\times3\times10^{8}}{390\times10^{-9}}\right] - \left[1.6\times10^{-19}\times1.10\right]$$

$$= 5.10\times10^{-19} - 1.76\times10^{-19} = 3.34\times10^{-19}\,J$$

$$= 2.09\ eV$$

ii) The threshold wavelength is

$$\lambda_0 = \frac{hc}{\phi_0} = \frac{6.626\times10^{-34}\times3\times10^{8}}{3.33\times10^{-19}}$$

$$= 5.969\times10^{-7}\,m = 5963\ \mathring{A}$$